

Due: Friday 15th November 2024, 11:59pm

Read the following paper, which presents a growth and remodeling theory to model adaptations to altered flow and pressure:

Valentin A, Cardamone L, Baek S, Humphrey JD. Complementary vasoactivity and matrix remodelling in arterial adaptations to altered flow and pressure. *Journal of The Royal Society Interface*. 2009 Mar 6;6(32):293-306.

Note: 1) Equation numbers correspond to equation numbers in the Valentin et al., RSIF, 2009. 2) Keep the symbols in your derivations consistent with symbols in Valentin et al., RSIF, 2009

1. Altered hemodynamic conditions

Experiments across multiple temporal and spatial scales suggest the existence of a mechanical homeostasis state in a blood vessel. Current research in vascular growth and remodeling is aimed at elucidating underlying mechanisms and predicting the final state of adaptation to perturbed hemodynamic conditions.

- (a) Notwithstanding the complexities due to geometry, pulsatile flow and anisotropic wall, it is instructive to consider idealized geometries and mean values to understand implications of altered hemodynamics. Consider an idealized cylindrical (thin vessel) geometry and derive the relationship between initial and final radius and thickness. Assume the state of stress in the vessel wall is a restored after hemodynamic perturbation and axial stress is not perturbed during adaptation. Predict the change in radius and thickness for the following increases in pressure and flow.
 - i. 300% increase in pressure
 - ii. 20% increase in flow
 - iii. 300% increase in pressure and 20% increase in flow
- (b) Download GnRadaptation.zip provided on the class website and read the comments in GnRadaptationvein.m for instructions on how to run simulations. The code requires three inputs – percentage increase in pressure, percentage increase in flow and output file name. Run the code for the following hemodynamic perturbations:
 - i. 300% increase in pressure
 - ii. 20% increase in flow
 - iii. 300% increase in pressure and 20% increase in flow
- (c) How do the adaptations compare against the predicted radius and thickness values?
- (d) For each of the above-mentioned perturbations, is the final radius and thickness same as the initial radius and thickness? Why?
- (e) Explain the radius and thickness evolution curves - why is there a spike in the curve initially?

2. Biaxial behaviour

The passive mechanical response of a vessel (no active smooth muscle stress) is usually determined by fitting the constituent constitutive equations to a biaxial experiment.

- (a) Using a simple rule of mixtures, derive the relationship between passive stress state in a vessel and the constituent strain energy density function for a mixture of structurally significant constituents - elastin, smooth muscle and 4-fiber collagen. Since biaxial tests are usually pressure - diameter experiments at different axial stretch derive the stress relations for just θ and z directions.
- (Hint: Start with individual strain energy density functions for elastin (equation 2.12) and collagen/smooth muscle (equation 2.13) and compute individual constituent derivatives with respect to stretch along θ and z directions. Don't forget to use the evolving configurations in your derivatives (equations 2.9 and 2.10)! Plug these derivatives into equation 2.4, using simple rule of mixtures, to get an expression for stress along θ and z direction. Because its a passive response, you can neglect the active stress term)*
- (b) Download fitstress8parameter.m from the class website. The function takes a vector of 8 passive constitutive parameters as inputs $[c_1^\theta(Pa), c_2^\theta, c_1^z(Pa), c_2^z, c_1^{diag}(Pa), c_2^{diag}, c^e(Pa), \alpha_0^{diag}]$. Run the code and produce biaxial curves for the following values:
- $[24000, 0.1, 1000, 0.05, 71.2, 1.3, 1000, 0.6388]$ - values for a vein
 - $[24000, 0.05, 1000, 0.05, 71.2, 1.3, 1000, 0.6388]$ - values for a hypothetical remodeled numerical vein
 - $[24000, 0.1, 1000, 0.05, 71.2, 1.1, 1000, 0.6388]$ - values for a hypotehtical remodeled numerical vein
- (c) Do you see a shift in the curve along a particular direction? Why?
- (d) Explain these results within the context of a venous adaptation (see Fig. 3 - Sassani, 2013, J. BioMech.) - how have the biaxial curves changed for a vein graft?